MIT Academy of Engineering, Alandi

Name : Ram Mahadev Pandhare
PRN : 202501040071
Div : CS1

Practical 3 : Numpy Library

3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

- The firstline contains two space-separated integers, and , representing the number of rows andthe number of columns.
- The nextlines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use reshape() function to reshape the input array with the specified number of rows and columns.

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CODE:
```
import numpy as np
rows, cols =map(int,input().split())

elements=[]
for _ in range(rows) :
    elements.extend(map(int,input().split()))

array= np.array(elements).reshape(rows,cols)

print(array)

print(array.ndim)

print(array.shape)

print(array.size)
```

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Average time	Maximum time
0.025 s	0.044 s
25.25 ms	44.00 ms

2 out of 2 shown test case(s) passe

10 out of 10 hidden test case(s) pas

Test case 1 44 ms Debug

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]	[[1 2 3 4]
[5 6 7 8]	[5 6 7 8]
[9 10 11 12]]	[9 10 11 12]]
2	2
(3, 4)	(3, 4)
12	12

Test case 2 11 ms Debug

Expected output	Actual output
0 0	0 0
[]	[]
2	2
(0, 0)	(0, 0)
0	0

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PRN : 202501040071

Lab Assignment

3.2.1. Numpy: Matrix Operations

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:
You are required to compute and display the results of the following matrix operations:

1. Addition ()
2. Subtraction ()
3. Element-wise Multiplication () 4.

Matrix Multiplication ()

5. Transpose of Matrix A

Input Format:

- The userwill input 3 rows for , each containing 3 integers separated by spaces.
- Similarly,the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:
The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication. 4.

The result of Matrix Multiplication.

5. The Transpose of Matrix A.

CODE:

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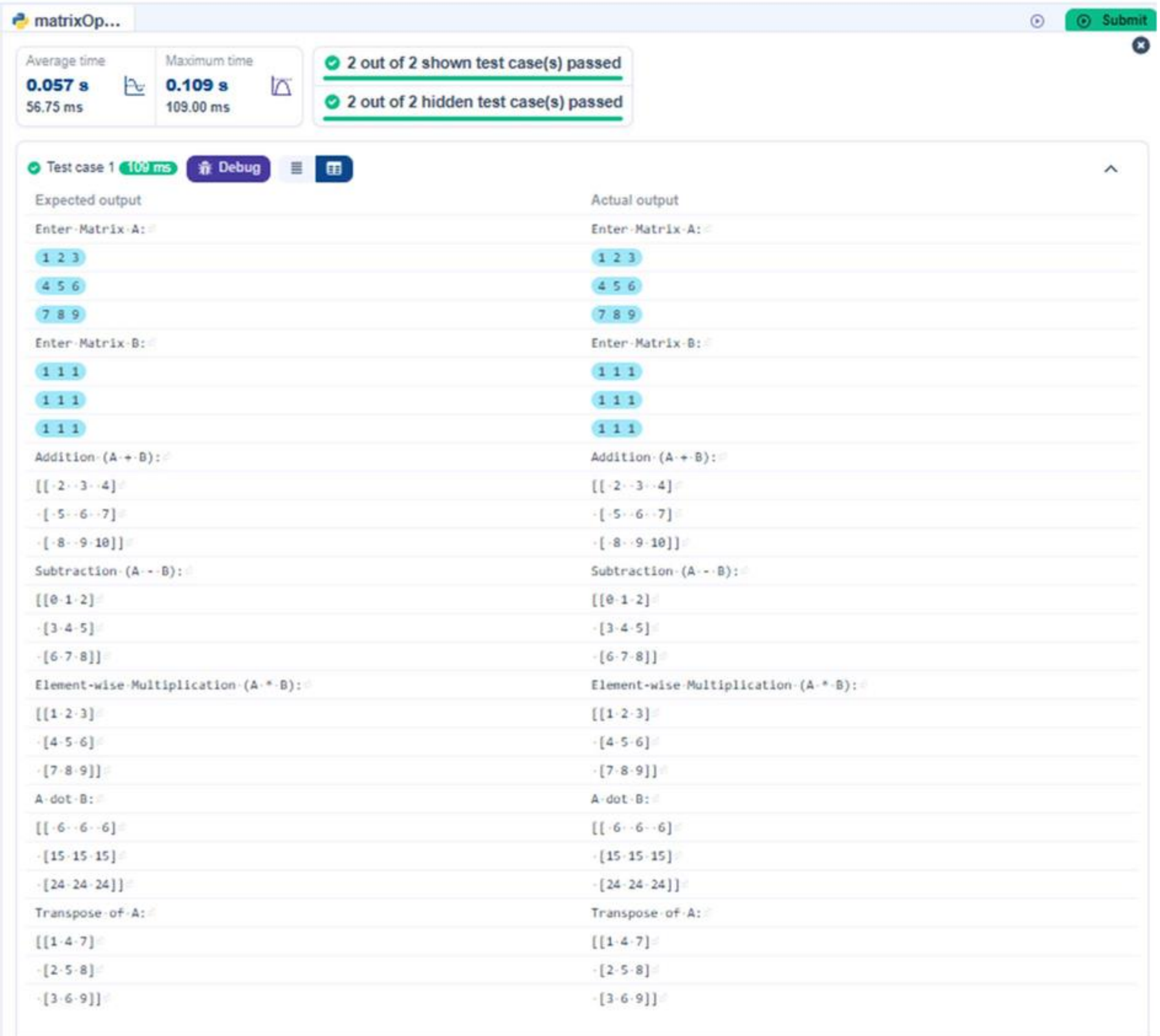
```
import numpy as np


#Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])



#Addition
addition_result = matrix_a + matrix_b
print("Addition (A + B):")
print(addition_result)
# Subtraction
substraction_result = matrix_a -
matrix_b print("Subtraction (A - B):")
print(substraction_result)

#Multiplication (element-wise)
elementwise_multiplication_result = matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result)



#Matrix multiplication (dot product)

matrix_multiplication_result =np.dot(matrix_a, matrix_b)
print("A dot B:")
```

```
print(matrix_multiplication_result)
# Transpose
transpose_a = matrix_a.T
print("Transpose of A:")
print(transpose_a)
```



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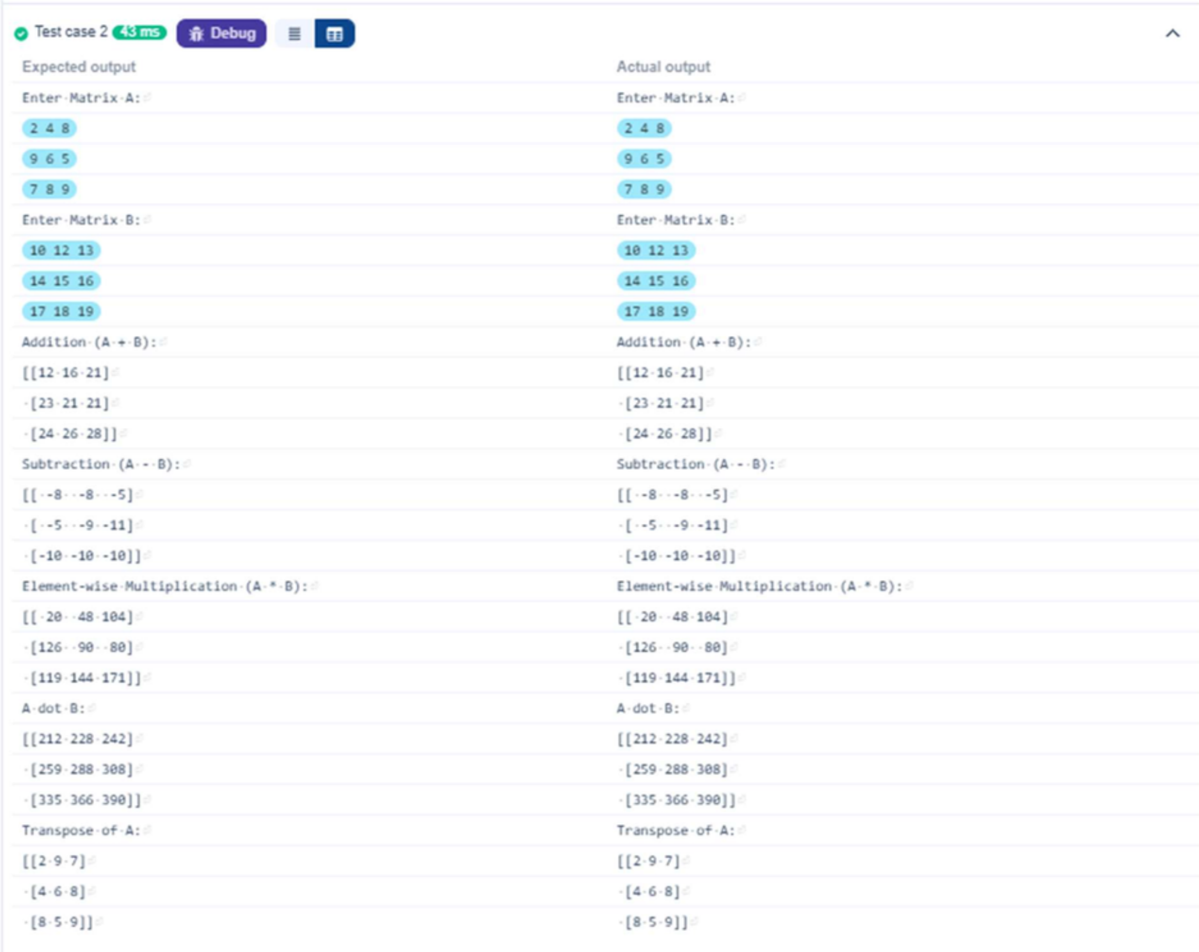

Test case 2 43 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
2 4 8	2 4 8
9 6 5	9 6 5
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
10 12 13	10 12 13
14 15 16	14 15 16
17 18 19	17 18 19
Addition (A + B):	Addition (A + B):
[[12 16 21]	[[12 16 21]
[23 21 21]	[23 21 21]
[24 26 28]]	[24 26 28]]
Subtraction (A - B):	Subtraction (A - B):
[[-8 -8 -5]	[[-8 -8 -5]
[-5 -9 -11]	[-5 -9 -11]
[-10 -10 -10]]	[-10 -10 -10]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[20 48 104]	[[20 48 104]
[126 90 80]	[126 90 80]
[119 144 171]]	[119 144 171]]
A dot B:	A dot B:
[[212 228 242]	[[212 228 242]
[259 288 308]	[259 288 308]
[335 366 390]]	[335 366 390]]
Transpose of A:	Transpose of A:
[[2 9 7]	[[2 9 7]
[4 6 8]	[4 6 8]
[8 5 9]]	[8 5 9]]

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

08:24

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking: Stack the two matrices horizontally (side by side).
- Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

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Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np

#Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

#Perform horizontal stacking (hstack)
a=np.hstack((arr1,arr2))
print("Horizontal Stack:")
print(a)

#Perform vertical stacking (vstack)
b=np.vstack((arr1,arr2))
print("Vertical Stack:")
print(b)
```

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Average time	Maximum time
0.052 s	0.066 s
52.00 ms	66.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 66 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6
7 8 9	7 8 9
10 11 12	10 11 12
Horizontal Stack:	Horizontal Stack:
[[1 2 3 4 5 6]	[[1 2 3 4 5 6]
[4 5 6 7 8 9]	[4 5 6 7 8 9]
[7 8 9 10 11 12]]	[7 8 9 10 11 12]]
Vertical Stack:	Vertical Stack:
[[1 2 3]	[[1 2 3]
[4 5 6]	[4 5 6]
[7 8 9]	[7 8 9]
[4 5 6]	[4 5 6]
[7 8 9]	[7 8 9]
[10 11 12]]	[10 11 12]]

Test case 2 45 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
-1 -2 -3	-1 -2 -3
-4 -5 -6	-4 -5 -6
-7 -8 -9	-7 -8 -9
Enter Array2:	Enter Array2:
10 11 12	10 11 12
13 14 15	13 14 15
16 17 18	16 17 18
Horizontal Stack:	Horizontal Stack:
[[-1 -2 -3 10 11 12]	[[-1 -2 -3 10 11 12]
[-4 -5 -6 13 14 15]	[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]]	[-7 -8 -9 16 17 18]]
Vertical Stack:	Vertical Stack:
[[-1 -2 -3]	[[-1 -2 -3]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[10 11 12]	[10 11 12]
[13 14 15]	[13 14 15]
[16 17 18]]	[16 17 18]]

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3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The userwill input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

```
import numpy as np

#Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())

#Generate the sequence using np.arange()
n=np.arange(start,stop, step)

#Print the generated sequence
print(n)
```

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

08:30

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. ArithmeticOperations :
   - Compute the element-wise sum, difference, and product of the two arrays.
2. StatisticalOperations :
   - Calculate the mean, median, and standard deviation of array A.
3. BitwiseOperations :
   - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: Ai OR Bi).

Input Format:

- The firstline contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

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- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```
import numpy as np

defarray_operations(A, B):

    # Convert A and B to NumPy arrays A
    = np.array(A)
    B = np.array(B)
    # Arithmetic Operations
    sum_result = A + B
    diff_result = A -B
    prod_result = A * B

    # Statistical Operations
    mean_A = np.mean(A)
    median_A = np.median(A)
    std_dev_A = np.std(A)

    # Bitwise Operations
    and_result = A & B
    or_result = A | B
    xor_result = A ^ B

  #Output results with one space between each element
    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
    print("Element-wise Difference:", ' '.join(map(str, diff_result)))
```

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```
        print("Element-wise Product:", ' '.join(map(str, prod_result)))


        print(f"Mean of A: {mean_A}")
        print(f"Median of A: {median_A}")
        print(f"Standard Deviation of A: {std_dev_A}")


        print("Bitwise AND:", ' '.join(map(str, and_result)))
        print("Bitwise OR:", ' '.join(map(str, or_result)))
        print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A=list(map(int, input().split())) # Elements of array A
B=list(map(int, input().split())) # Elements of array B
array_operations(A, B)
```

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3.2.5. Numpy: Copying and Viewing Arrays

04:23

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A singleline of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: < original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: < original_array>

Copy array: <copy_array>

```
import numpy as np

inputlist = list(map(int,input().split(" ")))

#Original array
original_array = np.array(inputlist)

#Create a view
```

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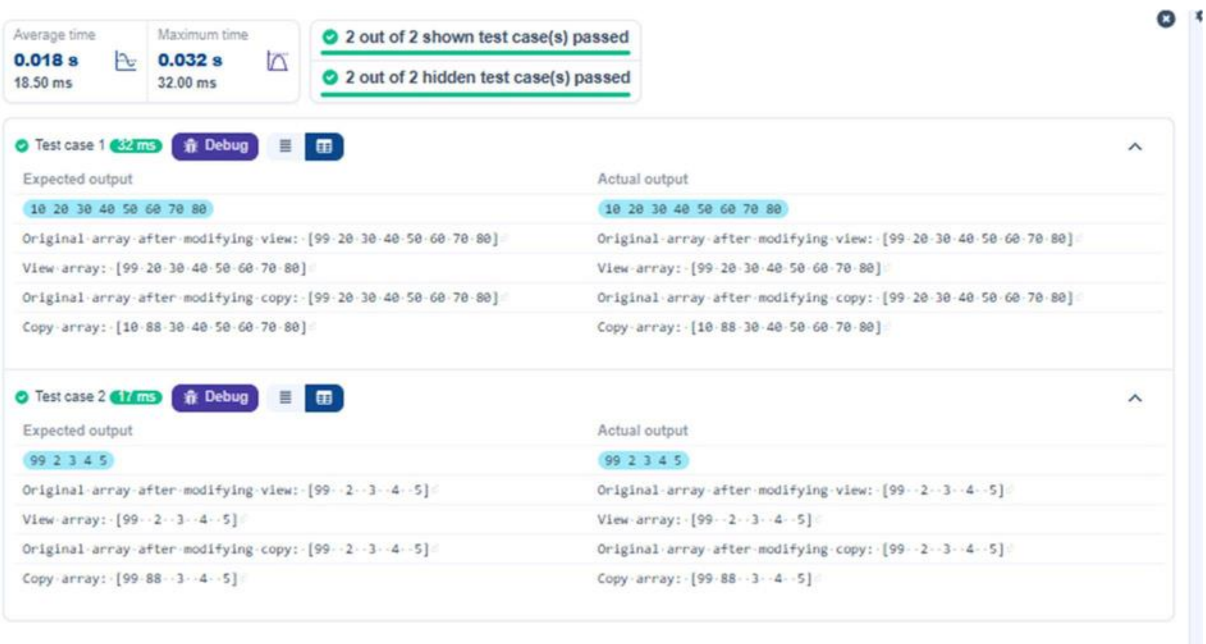
```
view_array =original_array.view()

#Create a copy
copy_array =original_array.copy()

#Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

#Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```



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3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

05:03

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. Searching: Find the indices where search_value appears in array1 and print these indices. 2.

Counting: Count how many times count_value appears in array1 and print the count.

3. Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.

4. Sorting: Sort array1 in ascending order and print the sorted array.

Input Format:

1. A singleline containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

Output Format:

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

```
import numpy as np
```

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```
#Input array from the user
array1 = np.array(list(map(int, input().split())))

#Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

#Find indices where value matches in array1
a=np.where(array1==search_value)[0] print(a)
# Count occurrences in array1
b=np.count_nonzero(array1==count_value)
print(b)
# Broadcasting addition
c=array1 + broadcast_value
print(c)
# Sort the first array
d=np.sort(array1)
print(d)
```

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Average time
0.035 s
34.75 ms

Maximum time
0.061 s
61.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 61 ms Debug

Expected output	Actual output
1 1 1 2 2 2	1 1 1 2 2 2
Value to search: 1	Value to search: 1
Value to count: 2	Value to count: 2
Value to add: 2	Value to add: 2
[0 1 2]	[0 1 2]
3	3
[3 3 3 4 4 4]	[3 3 3 4 4 4]
[1 1 1 2 2 2]	[1 1 1 2 2 2]

Test case 2 27 ms Debug

Expected output	Actual output
1 2 3 4 5	1 2 3 4 5
Value to search: 6	Value to search: 6
Value to count: 6	Value to count: 6
Value to add: 6	Value to add: 6
[]	[]
0	0
[7 8 9 10 11]	[7 8 9 10 11]
[1 2 3 4 5]	[1 2 3 4 5]

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3.2.7. Student Data Analysis and Operations

31:34

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.
- Find total students: Determine the total number of students in the dataset.
- Print all student roll numbers: Extract and print the roll numbers of all students.
- Print Subject 1 marks: Extract and print the marks of all students in Subject 1.
- Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.
- Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
- Print all subject marks: Display the marks of all students for each subject.
- Find total marks of students: Compute the total marks for each student across all subjects.
- Findthe average marks of each student : Compute the average marks for each student.
- Findaverage marks of each subject : Compute the average marks for all students in each subject.
- FindaveragemarksofSubject 1 and Subject 2 : Compute the average marks for Subject 1 and Subject 2.
- FindaveragemarksofSubject 1 and Subject 3 : Compute the average marks for Subject 1 and Subject 3.
- Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.
- Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.
- Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- Findthecountofstudentswhogotlessthan40marksinSubject 1: Count the number of students who scored less than 40 marks in Subject 1.
- Findthecountofstudentswhogotmorethan90marksinSubject 2 : Count the number of students who scored more than 90 marks in Subject 2.

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- Find the count of students who scored >=90 in each subject :Count the number of students who scored 90 or more marks in each subject.
- Find the count of subjects in which each student scored >=90: Determine how many subjects each student scored 90 or more marks in.
- PrintSubject1marksinascending order: Sort and print the marks of students in Subject 1 in ascending order.
- Print students who scored between 50 and 90 in Subject 1 :Display students who scored marks between 50 and 90 in Subject1.
- Find index positions of students who scored 79 in Subject 1: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

CODE:

```
import numpy as np

a= np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

#1. Print all student details
print("All student Details:\n", a )

#2. print total students

print("Total Students:",len(a) )

#3. Print all student Roll numbers
print("All Student Roll Nos", a[:,0]      )

#4. Print subject 1 marks
```

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print("Subject 1 Marks", a[:,1])

#5. print minimum marks of Subject 2

print("Min marks in Subject 2", np.min(a[:, 2]))

#6. print maximum marks of Subject 3

print("Max marks in Subject 3", np.max(a[:,3]))

#7. Print All subject marks

print("All subject marks:", a[:,1:])

#8. print Total marks of students print("Total

Marks", np.sum(a[:,1:], axis = 1))

#9. print average marks of each student

print(np.round(np.mean(a[:,1:], axis = 1),1))

#10. print average marks of each subject

print("Average Marks of each subject", np.mean(a[:, 1:], axis = 0))

#11. print average marks of S1 and S2

print("Average Marks of S1 and S2", np.mean(a[:, 1:3],axis=0))

#12. print average marks of S1 and S3

print("Average Marks of S1 and S3", np.mean(a[:, [1, 3]], axis = 0))

#13. print Roll number who got maximum marks in Subject 3 max_sub3_index

= np.argmax(a[:, 3])

print("Roll no who got maximum marks in Subject 3", a[max_sub3_index , 0])

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```
#14. print Roll number who got minimum marks in Subject 2 min_sub2_index
= np.argmin(a[:, 2])
print("Roll no who got minimum marks in Subject 2", a[min_sub2_index, 0]        )

#15. print Roll number who got 24 marks in Subject 2

print(f"Roll no who got 24 marks in Subject 2 [{a[a[:,2] == 24][:,0]}]"                    )

#16. print count of students who got marks in Subject 1 < 40
count_sub1_lt_40 = np.sum(a[:,1]<40)
print("Count of students who got marks in Subject 1 < 40", count_sub1_lt_40             )

#17. print count of students who got marks in Subject 2 > 90
count__sub2_gt_90 = np.sum(a[:,2]>90)
print("Count of students who got marks in Subject 2 > 90:", count__sub2_gt_90            )

#18. print count of students in each subject who got marks >= 90
count_each_subject_90plus = np.sum(a[:,1:]>=90 , axis = 0)
print("Count of students in each subject who got marks >= 90:", count_each_subject_90plus
)

#19. print count of subjects in which each student got marks >= 90
print("Roll no:", a[:,0].astype(float)        )
print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:] >= 90 , axis = 1)
)

#20. Print S1 marks in ascending order
print(np.sort(a[:,1]))
```

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#21. Print S1 marks >= 50 and <= 90

s1_filteredv = a[(a[:,1]>=50) & (a[:,1]<=90)]

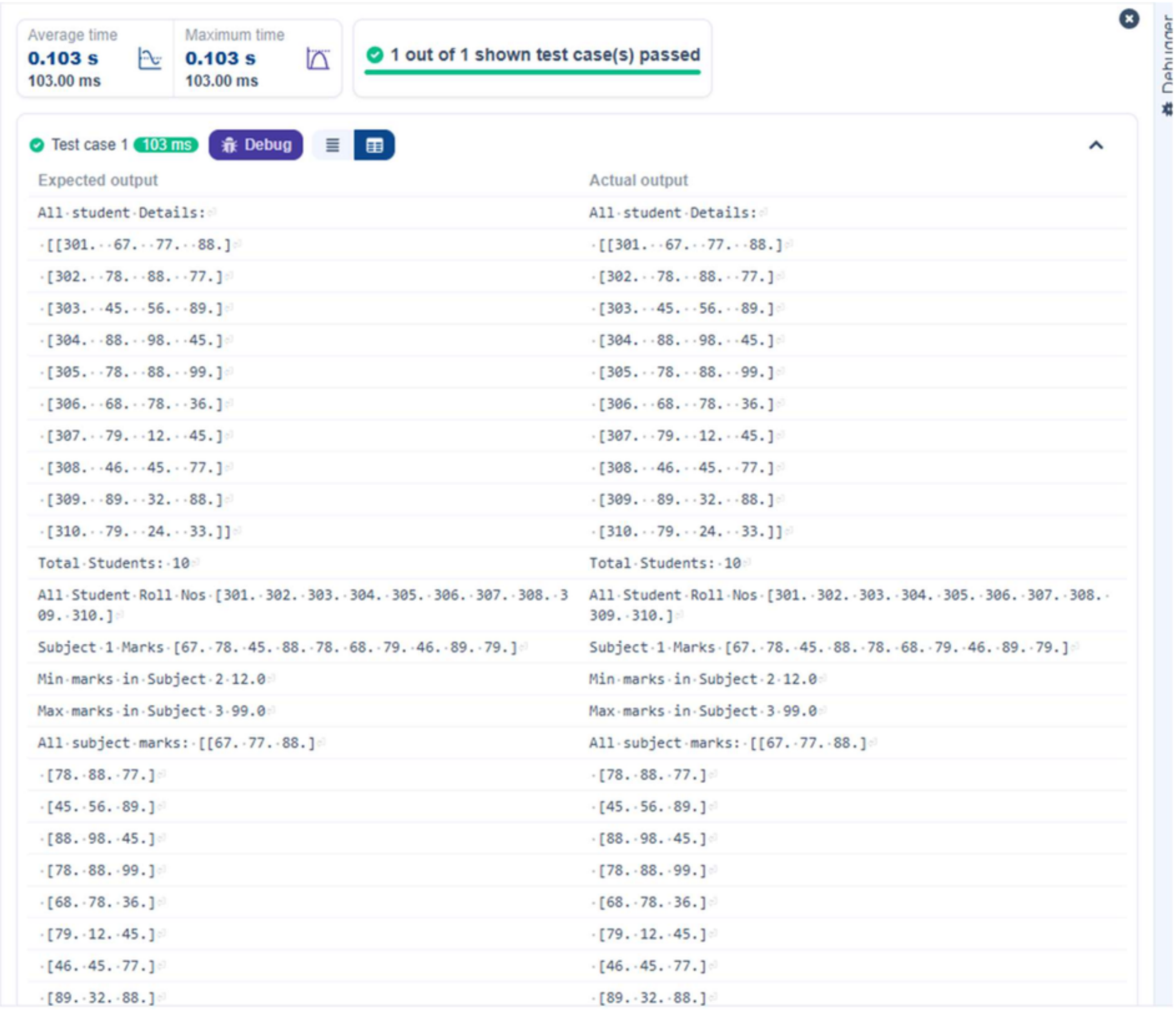
print(s1_filteredv)

print(a)

#22. Print the index position of marks 79

indices_79 = np.where(a[:,1:]==79)[0]

print((indices_79,))



Average time 0.103 s 103.00 ms | Maximum time 0.103 s 103.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 103 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]	Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]
Min marks in Subject 2 12.0	Min marks in Subject 2 12.0
Max marks in Subject 3 99.0	Max marks in Subject 3 99.0
All subject marks: [[67. 77. 88.]	All subject marks: [[67. 77. 88.]
[78. 88. 77.]	[78. 88. 77.]
[45. 56. 89.]	[45. 56. 89.]
[88. 98. 45.]	[88. 98. 45.]
[78. 88. 99.]	[78. 88. 99.]
[68. 78. 36.]	[68. 78. 36.]
[79. 12. 45.]	[79. 12. 45.]
[46. 45. 77.]	[46. 45. 77.]
[89. 32. 88.]	[89. 32. 88.]